Linear Photon Dispersion is Not a Prediction of Tetrahedral Emergent Gravity

Gradient expansion on a 4-valent network, Maxwell sector, and a consistency check against GRB 221009A

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Abstract

Tetrahedral Emergent Gravity (TEG) derives an effective vacuum dimension $D_{\text{eff}} = \ln 8$ and a roughness parameter $\sigma_{\text{eff}} = 0.1088$ from tetrahedral coordination $z_{\text{fund}} = 4$ together with orientation duality on network links. Those ingredients do not automatically supply a linear modification of the photon dispersion relation. This note records a restricted but explicit argument: on a 4-valent network with linkwise orientation duality and residual S_3 isotropy at the node, the gradient expansion of the graph d'Alembertian contains no odd powers of $\ell\partial$. The first kinematic correction, if present, is of order $n = 2$. The same parity cancellation applies to the discrete Hodge Laplacian acting on a one-form, so a Myers–Pospelov birefringent splitting at $\mathcal{O}(E/E_{\text{Pl}})$ is likewise not generated. A linear ansatz $\xi_1 = \ln 8/(2\pi)$ is therefore external to the TEG axiom set. Evaluated on GRB 221009A ($z = 0.151$) at $E = 12.2$ TeV it predicts a time delay $\Delta t \approx 24$ s, already in tension with the LHAASO bound $E_{\text{QG},1} > 10E_{\text{Pl}}$ ($\xi_1 < 0.1$). The operative TEG statement in the photon sector is $\xi_1 = \xi_b = 0$ at linear order. This is a consistency constraint, not an empirical confirmation of $z_{\text{fund}} = 4$. The model continues to be tested where it has a derived equation: galactic rotation curves and the holographic codimension $\partial = 3 - \ln 8$.

Contents

1 Purpose and epistemic status

This document is a technical note, not a replacement for the TEG papers [1–3]. Its only aims are:

1. to state what TEG does not predict in the photon sector;
2. to keep that statement aligned with symmetries already used to derive $D_{\mathrm{eff}} = \ln 8$ and σ_{eff};
3. to confront one concrete linear ansatz with published time-of-flight and polarimetry bounds, using a real event rather than a schematic source at 100 Mpc.

Status. The lemmas below are continuum-limit arguments on a graph operator. They are not a derivation from the EPRL intertwiner or from a completed Maxwell action on the quaternionic vacuum. Open problems remain labelled as such. Numerical delays use a standard flat-ΛCDM integral; they are reproducible and are not fitted.

No claim is made that the absence of linear photon LIV supports the tetrahedral axiom. Many unrelated theories also have $\xi_1 = 0$. The useful content is negative: a linear delay of $\mathcal{O}(1) \times E/E_{\mathrm{Pl}}$ is not a TEG output and should not be advertised as one.

2 What TEG has actually derived

The working axiom of TEG is that the quantum vacuum in $\mathbb{R}^3$ selects tetrahedral coordination $z_{\mathrm{fund}} = 4$ [1]. Orientation duality on each link (bidirectional information flow) promotes this to an effective spectral dimension

$$d_s = \ln(2z_{\mathrm{fund}}) = \ln 8 \approx 2.079. \tag{1}$$

The same chain gives $N_{\mathrm{bits}} = 3$ and $\sigma_{\mathrm{eff}} = \sigma_{\mathrm{UV}}/3 \approx 0.1088$ used without per-galaxy fitting on the SPARC sample [1,4]. The holographic codimension is

$$\partial = 3 - \ln 8 \approx 0.9206. \tag{2}$$

Later working drafts interpret $z_{\text{fund}} = 4$ as the projection of a 5-cell in S^3 [3]. That construction identifies the lost bit $\ln 2$ with a gravitational shadow. It does not identify it with a drift of the electromagnetic phase velocity.

These objects describe the support of the vacuum network (diffusion, spectral dimension, roughness). They do not by themselves specify a correction

$$E^2 = p^2c^2 \left[1 \pm \xi_1 \frac{E}{E_{\text{Pl}}}\right]. \tag{3}$$

Equation (3) is a separate kinematic ansatz. One might be tempted to identify the single dimensionless number $\ln 8$ from the vacuum geometry with the LIV coefficient; inserting $\xi_1 = \ln 8/(2\pi)$ couples a TEG number to a Maxwell operator that has not been derived from the axiom.

3 Gradient expansion of the graph wave operator

3.1 Network hypotheses

At a node the four outgoing unit vectors $\mathbf{n}_i \in \mathbb{R}^3$ satisfy tetrahedral closure

$$\sum_{i=1}^{4} \mathbf{n}_i = \mathbf{0}. \tag{4}$$

Orientation duality is implemented as a $\mathbb{Z}_2$ identification on every link,

$$\mathbf{n}_i \longleftrightarrow -\mathbf{n}_i, \tag{5}$$

with equal weight on both orientations. This is the same doubling $z \to 2z$ that produces Eq. (1). A residual S_3 action on the three independent spatial directions at the node is the same symmetry used in TEG to force $\sigma_{\text{eff}} = \sigma_{\text{UV}}/3$.

A test scalar (standing in for the principal symbol of the wave operator) obeys

$$(\Box_{\text{net}}\phi)(x) = \frac{1}{\ell^2} \sum_{i=1}^{4} \sum_{\varepsilon=\pm} \big[\phi(x + \varepsilon\ell\mathbf{n}_i) - \phi(x)\big] - \frac{1}{c^2}\partial_t^2\phi, \tag{6}$$

where ℓ is the network spacing.

3.2 Cancellation of odd orders

Taylor expansion of each shift gives

$$\begin{aligned}\phi(x + \varepsilon\ell\mathbf{n}) = \phi + \varepsilon\ell n^a\partial_a\phi + \frac{\ell^2}{2}n^a n^b\partial_a\partial_b\phi + \frac{\varepsilon\ell^3}{6}n^a n^b n^c\partial_a\partial_b\partial_c\phi \\ + \frac{\ell^4}{24}n^a n^b n^c n^d\partial_a\partial_b\partial_c\partial_d\phi + O(\ell^5).\end{aligned} \tag{7}$$

Summing $\varepsilon = \pm$ eliminates every term odd in ε:

$$\sum_{\varepsilon=\pm} \varepsilon = 0, \qquad \sum_{\varepsilon=\pm} 1 = 2. \tag{8}$$

The cubic piece $\sim \ell^3\partial^3$, which would generate linear LIV of type (3), is absent already at a single link. After summing the four links and imposing Eq. (4) together with S_3 isotropy, one obtains

$$\sum_{i=1}^{4} n_i^a n_i^b = \frac{4}{3}\delta^{ab}, \tag{9}$$

and therefore

$$\Box_{\rm net} = \Box_{\rm Mink} + \kappa \ell^2 \Delta^2 + O(\ell^4), \tag{10}$$

with κ a pure number of order one fixed by the tetrahedron and not needed here.

Lemma 1 (No linear scalar LIV)**.** *On a 4-valent network with tetrahedral closure, linkwise orientation duality, and residual S_3 isotropy, the gradient expansion of $\Box_{\rm net}$ contains no odd powers of $\ell\partial$. In particular $\xi_1 = 0$. The first allowed kinematic correction is $n = 2$.*

Remark 1. Lemma 1 uses no symmetry that TEG does not already invoke for d_s and $\sigma_{\rm eff}$. A hand-inserted odd operator that produces $\xi_1 = \ln 8/(2\pi)$ would break the orientation duality that defines Eq. (1).

3.3 What is not claimed

Lemma 1 does not compute ξ_2 from EPRL data. It does not address a fractional Calcagni measure; such a measure changes volumes and need not reintroduce $\ell^3\partial^3$ if the network remains $\mathbb{Z}_2$-dual [5,6]. It is a statement about the graph operator (6), not a completed proof in spin-foam perturbation theory.

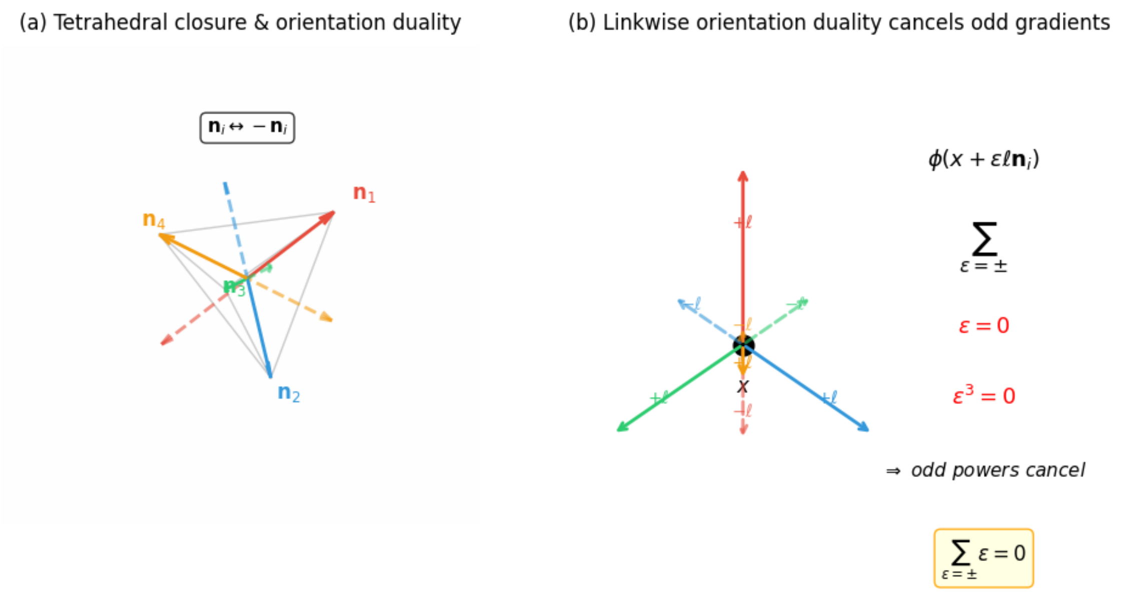

Figure 1: (a) Tetrahedral closure: the four unit vectors $\mathbf{n}_i$ from a node sum to zero. Orientation duality (solid vs. dashed arrows) assigns equal weight to $\mathbf{n}_i$ and $-\mathbf{n}_i$ on every link. (b) At each link the Taylor expansion contains terms odd in ε; the sum $\sum_{\varepsilon=\pm}$ forces their cancellation. Consequently the graph d'Alembertian has no linear-order Lorentz-violating term.

4 Maxwell sector and vacuum birefringence

Vacuum birefringence of Myers–Pospelov type [7] assigns opposite linear shifts to the two photon helicities,

$$E_\pm^2 = p^2c^2 \left[1 \pm \frac{2\xi_b}{E_{\rm Pl}}\,E\right]. \tag{11}$$

Linear polarisation then rotates by an energy-dependent angle

$$\Delta\theta \simeq \xi_b \frac{E^2}{E_{\rm Pl}} \int_0^z \frac{(1+z')}{H(z')}\,dz'. \tag{12}$$

This is a different observable from a common time delay ξ_1.

On the tetrahedral complex the potential is a discrete one-form A_e on oriented edges, with

$$A_{-e} = -A_e. \tag{13}$$

Equation (13) is orientation duality; it is not an extra assumption. The curvature lives on faces, $F_f = (\mathrm{d}A)_f$ and the wave equation is the discrete Hodge Laplacian $(\delta\mathrm{d} + \mathrm{d}\delta)A = 0$. In the continuum projection one may write

$$(\Box_{\mathrm{net}} A)^a = \frac{1}{\ell^2}\sum_{i=1}^{4}\sum_{\varepsilon=\pm} P^{ab}(\mathbf{n}_i)\Big[A_b(x+\varepsilon\ell\mathbf{n}_i) - A_b(x)\Big] - \frac{1}{c^2}\partial_t^2 A^a, \tag{14}$$

where $P^{ab}(\mathbf{n}) = \delta^{ab} - n^a n^b$ removes the longitudinal component along the link.

The same $\varepsilon = \pm$ sum cancels odd gradient orders. A helicity-splitting operator at $\mathcal{O}(\ell\partial)$ has to be odd under $\mathbf{n} \to -\mathbf{n}$ (a lattice Chern–Simons or Carroll–Field–Jackiw structure). That object changes sign with orientation duality and vanishes after the sum. S_3 further removes any leftover director u^μ that could support $u_\mu \widetilde{F}^{\mu\nu} F_{\nu\rho}$ at the node: a regular tetrahedron has no S_3-invariant preferred vector.

Lemma 2 (No linear birefringence)**.** *On the same network, with $A_{-e} = -A_e$, the discrete Hodge Laplacian does not generate a Myers–Pospelov term. Both helicities share the same principal symbol at $\mathcal{O}(E/E_{\mathrm{Pl}})$, i.e. $\xi_b = 0$.*

An $n = 2$ correction common to both helicities remains allowed by parity. After S_3 averaging it is isotropic, so it does not split polarizations. Its time delay at TeV energies is many orders of magnitude below current timing resolution (Section 5).

Existing GRB polarimetry already constrains an $\mathcal{O}(1)$ coefficient in Eq. (**??**) at the level $|\xi_b| \lesssim 10^{-16}$ under standard assumptions [12–15]. TEG does not produce such a coefficient. An ansatz $\xi_b = \ln 8/(2\pi)$ is excluded by polarimetry independently of Lemma 2; the lemma only states that TEG does not ask for it.

5 Numerical consistency check

5.1 Convention

For the linear relation (3) the group-velocity delay is taken in the Jacob–Piran form [8]

$$\Delta t = \xi_1 \frac{E}{E_{\mathrm{Pl}}} \int_0^z \frac{(1+z')}{H(z')}\, dz', \qquad H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \tag{15}$$

Planck 2018 parameters $H_0 = 67.4$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$ are used as baseline; a TEG-like background $H_0 = 70.3$, $\Omega_m = 0.319$ changes the integrals by about four percent and does not affect the conclusion. The Planck energy is $E_{\mathrm{Pl}} = 1.2209 \times 10^{19}$ GeV. The integral at $z = 0.151$ evaluates to $I_1 = 7.157 \times 10^{16}$ s (Planck cosmology), corresponding to $D_L \approx 745$ Mpc.

The schematic Euclidean product $(E/E_{\mathrm{Pl}})(D/c)$ at $D = 100$ Mpc and $E = 100$ TeV equals 84.3 s for $\xi_1 = 1$. That number is not the prediction of $\xi_1 = \ln 8/(2\pi)$ and is not the delay on GRB 221009A.

5.2 Anchor event

GRB 221009A at $z = 0.151$ [9] supplies both a high-statistics TeV afterglow and a reported 12.2 TeV photon. LHAASO places a 95% lower limit $E_{\mathrm{QG},1} > 10E_{\mathrm{Pl}}$ on linear time-of-flight LIV from that afterglow [10], equivalent to $|\xi_1| < 0.1$ in the convention (15). HAWC observations of photons above 100 TeV independently exclude $\mathcal{O}(1)$ superluminal linear LIV through photon decay [11].

5.3 Results

Let $\xi_1^{(A)} = \ln 8/(2\pi) \approx 0.331$. Table 1 lists Δt from Eq. (15).

Table 1: Linear time delays on GRB 221009A ($z = 0.151$), Planck cosmology. The column $\xi_1 = 0.1$ is the LHAASO linear bound, not a TEG output.

Photon energy	$\xi_1 = 0.331$ (ansatz A)	$\xi_1 = 0.1$ (LHAASO cap)	$\xi_1 = 1$
12.2 TeV	23.7 s	7.2 s	71.5 s
99.3 GeV	0.19 s	0.06 s	0.58 s
100 TeV (if unattenuated)	194 s	59 s	586 s

Ansatz A overshoots the LHAASO cap at 12.2 TeV by a factor ≈ 3.3. It is therefore not a viable TEG prediction. The same conclusion holds, *a fortiori*, for $\xi_1 = 1$. A quadratic correction with $\xi_2 = \mathcal{O}(1)$ yields $\Delta t_{n=2} \sim 10^{-13}$ s at 12.2 TeV and is not an observational target.

5.4 Three mutually exclusive statements

A. $\xi_1 = \ln 8/(2\pi)$. External ansatz. In tension with LHAASO on this burst.

B. $\xi_1 = 0$ at linear order, as required by Lemmas 1 and 2. Compatible with LHAASO and HAWC.

C. Leading correction $n = 2$ with $\xi_2 = \mathcal{O}(1)$. Allowed by parity; unobservable with present timing.

The TEG axiom set selects B (and permits C). It does not select A.

6 Relation to the tetrahedral axiom

Lemma 1 does not increase the empirical probability of $z_{\text{fund}} = 4$. A null linear-LIV result is shared by Maxwell theory in Minkowski space and by many discrete models. What the lemmas do provide is a consistency filter: if $z_{\text{fund}} = 4$ plus orientation duality had forced $\xi_1 = \mathcal{O}(1)$, the framework would already be excluded by GRB 221009A. Because they force $\xi_1 = 0$, the photon sector does not kill the axiom. That is a necessary condition for taking the axiom seriously. It is not a sufficient condition for believing it.

The axiom continues to be tested where TEG writes an equation that depends on z_{fund}: the SPARC z-sweep [1,4], and the cosmological predictions attached to $\partial = 3 - \ln 8$ (weak-lensing shear and void density contrast) [2]. Those tests are independent of this note.

7 Conclusions

1. Linear photon LIV of the form (3) is not derived from the TEG axiom $z_{\text{fund}} = 4$.

2. The same orientation duality that defines $d_s = \ln 8$ cancels odd terms in the graph wave operator (Lemma 1) and in the discrete Maxwell operator (Lemma 2).

3. The working TEG statement is $\xi_1 = \xi_b = 0$ at $\mathcal{O}(E/E_{\text{Pl}})$.

4. The numerical ansatz $\xi_1 = \ln 8/(2\pi)$ predicts $\Delta t \approx 24$ s for a 12.2 TeV photon from GRB 221009A and is in tension with the published LHAASO linear bound.

5. A quadratic correction remains formally allowed and is phenomenologically inert at current sensitivity.

6. This note does not confirm tetrahedral coordination. It records that TEG need not fight existing photon-propagation data in order to keep its axiom.

Data and reproducibility

The integrals $I_1(z)$ and $I_2(z)$ are ordinary one-dimensional quadratures of Eq. (15). No SPARC or LHAASO microdata were reprocessed. Source versions of TEG remain those cited below. This note does not modify $\sigma_{\rm eff}$ or the galactic sector.

Acknowledgements

SPARC rotation-curve data [4] and the LHAASO analyses of GRB 221009A [9,10] are public. No new observations are reported here.

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